Md Kamruzzaman

AI & Python Backend Engineer | FastAPI | GenAI (RAG) | MLOps

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

AI & Python Backend Engineer with a B.Tech in Computer Science (Machine Learning specialization). Hands-on experience building production-ready REST APIs, deploying end-to-end ML systems, and developing Generative AI (RAG) pipelines. Skilled in FastAPI, LangChain, MLflow, Docker, and PostgreSQL. Focused on bridging the gap between scalable backend software engineering and modern AI infrastructure, with a long-term goal of becoming an MLOps/ML Platform Engineer.

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, REST APIs, SQLAlchemy, Alembic, JWT Authentication, Pydantic
Generative AI: LangChain, RAG Pipelines, ChromaDB, Google Gemini API, Prompt Engineering
Machine Learning: scikit-learn, Logistic Regression, Gaussian Naive Bayes, TF-IDF, NLP, classification
MLOps & Monitoring: MLflow, Prometheus, Grafana, Evidently (Drift Detection), Model Serving
DevOps & Databases: Docker, Kubernetes (basics), CI/CD (GitHub Actions), Git, PostgreSQL, SQL

PROJECTS

Expense Tracker API — Backend Engineering & Deployment Code | Live API

- Built and deployed a secure RESTful API with JWT authentication, bcrypt hashing, and user-scoped data isolation.
- Designed a normalized PostgreSQL schema with SQLAlchemy ORM and Alembic migrations; implemented Pydantic validation, pagination, and structured error handling.
- Achieved 91% test coverage with pytest across authentication and CRUD flows; automated testing and Ruff linting via GitHub Actions CI/CD.

ML Expense Platform — MLOps & Model Serving Code | Live API

- Architected an end-to-end ML system that classifies expense categories from free-text descriptions.
- Trained a TF-IDF and Logistic Regression pipeline in scikit-learn; tracked and compared four MLflow experiments.
- Served the model via a FastAPI endpoint with Pydantic validation and lazy model loading for efficient inference.
- Instrumented Prometheus metrics for prediction counts and latency; visualized performance in Grafana dashboards.
- Implemented Evidently-based data drift detection, validating it by injecting simulated production data.
- Containerized with Docker and authored Kubernetes manifests with liveness/readiness probes for self-healing.

Docu-RAG API — Generative AI & LLM Integration Code

- Built a production-ready Retrieval-Augmented Generation (RAG) API using FastAPI and LangChain to enable context-aware question answering over custom PDF documents.
- Designed an ingestion pipeline that chunks PDFs and stores vector embeddings in ChromaDB.
- Implemented semantic search to retrieve relevant document chunks, which are then passed to the Google Gemini LLM to generate accurate, non-hallucinated answers.

EDUCATION

Lovely Professional University, Punjab, India 2025
B.Tech in Computer Science and Engineering, Machine Learning Specialization

CERTIFICATIONS

- Natural Language Processing Specialization — DeepLearning.AI, Coursera
- Meta Back-End Developer — Coursera
- Python Data Structures — University of Michigan, Coursera